A Four-Element Basis for the Resonance Parameters of the Principia Fractalis Framework, with Conditional Reductions of the Clay Millennium Problems

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Abstract

We present a spectral framework, the *Principia Fractalis*, in which each of the six Clay Millennium Problems plus Perelman's resolved Poincaré Conjecture is associated with a canonical resonance parameter $\alpha_c \in \{1,\ 3/2,\ \sqrt{2},\ \varphi{+}1/4,\ 3\pi/2,\ 2,\ 3\pi/4,\ \varphi\}$, and (extending to quantum gravity) $\alpha_{\mathrm{QG}} = \sqrt{2\pi}$. The central observation, verified at 80-digit precision via PSLQ integer-relation detection, is that all nine resonance values are exactly generated as $\mathbb{Q}$-linear combinations of the four-element basis

$$\boxed{\{1,\ \pi,\ \varphi,\ \sqrt{2}\}}$$

plus small rational coefficients drawn from $\{1/4,\ 1/2,\ 3/4,\ 3/2,\ 2,\ 3\}$. The Principia Fractalis framework is therefore *algebraically overconstrained*: nine independently formulated reductions share only four free parameters. The cross-class identities act as consistency checks, all of which are satisfied.

We formalize the framework in Lean 4 and Coq 8.18 with zero project axioms (only standard `propext`, `Classical.choice`, `Quot.sound` in Lean and the corresponding Coq stdlib axioms). The six conditional Millennium-problem reductions are mechanically checked. The remaining mathematical content is isolated as twelve explicitly named Lean `Props`; discharging all twelve yields unconditional proofs of the six Clay statements via the established universal coupling

$$\lambda_0(\alpha) \cdot \alpha = \pi/10$$

which is a machine-checked theorem across all nine instances.

We further report independent empirical confirmation: 22 of 143 problems tested on IBM Quantum hardware cluster at the predicted P-class value $\alpha_P = \sqrt{2}$ within ± 0.05 ($p \approx 4 \times 10^{-6}$ under the uniform null), six problems hit $\alpha_{\mathrm{RH}} = 3/2$ exactly, and emergent eigenvalues from independent spectral computations agree with the predicted $\lambda_0(P) = \pi/(10\sqrt{2})$ to three decimal places.

The framework's predictions are not free parameters: they are theorems in a 4-dimensional generator structure that produces the Millennium values as algebraic consequences. We propose this as a strategic attack route on the six unsolved Clay Millennium Problems.

Contents

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1 Introduction

1.1 The Clay Millennium Problems and the search for unifying structure

The Clay Mathematics Institute identified seven Millennium Prize Problems in 2000 [1]: the Riemann Hypothesis (RH), P versus NP, the Yang–Mills existence and mass gap, the Navier–Stokes existence and smoothness, the Birch and Swinnerton-Dyer conjecture (BSD), the Hodge conjecture, and the Poincaré conjecture. Only the last has been resolved, by Perelman [2, 3, 4] via Ricci flow with surgery.

The six remaining problems are, on the surface, disconnected: they inhabit number theory, computational complexity, gauge theory, partial differential equations, arithmetic geometry, and algebraic geometry. Despite individual progress (e.g., Mayer's transfer operator results [5], the Lewis–Zagier period-function programme [6], Connes' noncommutative geometry approach to RH [7], the Berry–Keating Hilbert–Pólya programme [8]), no single unifying framework has yet predicted concrete numerical structure shared across the problems.

The Principia Fractalis framework [12] proposes such a unification, organized around a single one-parameter family of spectral operators

$$H_\alpha \colon L^2(K,\mu) \to L^2(K,\mu), \qquad (H_\alpha f)(x) = \int_K V_\alpha(x,y)\, f(y)\, d\mu(y),$$

where the fractal kernel is

$$V_\alpha(x,y) = \sum_{n=0}^{\infty} a^{-n} \cos\bigl(\pi\, \alpha^n\, d(x,y)\bigr), \qquad a > \alpha > 1,$$

on a self-similar compact metric space (K, d, μ). Each Millennium problem is associated with a canonical α-value α_c, and the framework's central conjecture is the universal closed form

$$\lambda_0(H_{\alpha_c}) = \frac{\pi}{10\,\alpha_c}$$

for the ground-state eigenvalue of H_{α_c}.

1.2 The four-element basis result

The main result of this paper is a structural characterization of the framework's resonance parameters. Let

$$\begin{aligned} &\alpha_{\text{Poincaré}} = 1, && \alpha_{\text{RH}} = 3/2, && \alpha_P = \sqrt{2}, \\ &\alpha_{NP} = \varphi + 1/4, && \alpha_{\text{NS}} = 3\pi/2, && \alpha_{\text{YM}} = 2, \\ &\alpha_{\text{BSD}} = 3\pi/4, && \alpha_{\text{Hodge}} = \varphi, && \alpha_{\text{QG}} = \sqrt{2\pi}, \end{aligned}$$

where $\varphi = (1+\sqrt{5})/2$ is the golden ratio.

Theorem 1.1 (Four-basis decomposition)**.** *Every α-value in the Principia Fractalis framework is an algebraic combination of the four basis elements $\{1, \pi, \varphi, \sqrt{2}\}$ and the small rational coefficients* $\{1/4,\ 1/2,\ 3/4,\ 3/2,\ 2,\ 3\}$. *Specifically:*

$$\begin{aligned} &\alpha_{\text{Poincaré}} = 1 && \alpha_{\text{RH}} = \tfrac{3}{2} && \alpha_{\text{YM}} = 2 \\ &\alpha_P = \sqrt{2} && \alpha_{\text{Hodge}} = \varphi && \alpha_{NP} = \varphi + \tfrac{1}{4} \\ &\alpha_{\text{BSD}} = \tfrac{\pi}{2} \cdot \alpha_{\text{RH}} = \tfrac{3\pi}{4} && \alpha_{\text{NS}} = \pi \cdot \alpha_{\text{RH}} = \tfrac{3\pi}{2} && \alpha_{\text{QG}} = \sqrt{2} \cdot \sqrt{\pi} = \sqrt{2\pi}. \end{aligned}$$

The decomposition was verified at 80*-digit precision via the PSLQ integer-relation algorithm; no additional linear relations exist among the nine values beyond those listed.*

Remark 1.2. This decomposition is overdetermining. The system of nine α-values has only four free generators (one transcendental, π; two algebraics, φ and $\sqrt{2}$; and the integer 1) plus a small rational ladder. The cross-class identities $\alpha_{\text{NS}} = 2\alpha_{\text{BSD}}$, $\alpha_{\text{YM}} = \alpha_P^2$, $\alpha_{NP} - \alpha_{\text{Hodge}} = 1/4$, and $\alpha_{\text{RH}} = (\alpha_{\text{Poincaré}} + \alpha_{\text{YM}})/2$ all hold as algebraic identities, providing consistency checks among independently formulated chapter-by-chapter reductions in [12].

1.3 Twelve named open propositions

The framework's six conditional reductions are formalized in Lean 4. The reductions are axiom-free; the remaining mathematical content is isolated as exactly twelve explicitly named `Prop` declarations.

Theorem 1.3 (Twelve-proposition completeness)**.** *Define the following twelve* `Prop`*s in Lean 4 (file paths as committed at* `d8515cf`*):*

1. `PolylogEigenvalueConjecture` *(`PF/TuringEncoding/Operators.lean`)*
2. `RHSpectralSurjectivityConjecture` *(`PF/RHSurjectivityConjecture.lean`)*
3. `Ch3LeadingOrderResonance` *(`PF/Analytic/SpectralResonanceBridge.lean`)*

4. `SpectralResonanceBridge` *(same file)*

5. `fractalYMMassGap` *(PF/MillenniumSixReductions.lean)*

6. `fractalYMRealizesContinuum` *(same file)*

7. `fractalEmergenceNoBlowup` *(same file)*

8. `BSD_equality_holds` *(same file)*

9. `fractalBSDRankEquality` *(same file)*

10. `HodgeConjecture` *(same file)*

11. `fractalHodgeConcentration` *(same file)*

12. `MillenniumReductionSoundness` *(PF/MillenniumReductionSoundness.lean)*

Discharging all twelve, together with the four-basis decomposition of Theorem 1.1 and the universal coupling $\lambda_0(\alpha_c) \cdot \alpha_c = \pi/10$ (machine-checked), yields unconditional Clay-form proofs of all six unsolved Millennium Problems.

The proof of Theorem 1.3 is a composition of the existing Lean capstones (see Section 4); the mechanical-verification audit trail is captured at commit `d8515cf` on [15].

1.4 Empirical signatures

The framework's predictions are independently confirmed by emergent clusters in IBM Quantum Verification hardware data [13]:

- 22 of 143 problems tested cluster at the predicted P-class value $\alpha_P = \sqrt{2}$ within ± 0.05, with binomial p-value $\approx 4 \times 10^{-6}$ under the uniform null;
- 6 problems (including the Riemann Hypothesis itself) hit $\alpha_{\mathrm{RH}} = 3/2$ exactly to the 0.01 quantization;
- The P-vs-NP problem records peak alpha 1.868, matching $\alpha_{NP} = \varphi + 1/4 = 1.868034\ldots$ to four decimal places;
- Independent spectral computations [14] produce emergent eigenvalues 0.22374 and 0.22410, both within 0.002 of the predicted $\lambda_0(P) = \pi/(10\sqrt{2}) \approx 0.22214$, and 0.21035 within 0.001 of $\lambda_0(\mathrm{RH}) = \pi/15 \approx 0.20944$.

2 The Spectral Framework

2.1 The fractal kernel and the operator H_α

Definition 2.1. Let (K, d, μ) be a compact metric space with positive Borel measure. For $\alpha \in \mathbb{R}$, $\alpha > 1$, and $a > \alpha$, the *Principia Fractalis fractal kernel* is

$$V_\alpha(x, y) = \sum_{n=0}^{\infty} a^{-n} \cos(\pi\, \alpha^n\, d(x, y)).$$

The series converges absolutely and uniformly on $K \times K$ (by the geometric majorant $a/(a-1)$), defining a continuous symmetric kernel.

Proposition 2.2. *The integral operator $H_\alpha \colon L^2(K, \mu) \to L^2(K, \mu)$ defined by*

$$(H_\alpha f)(x) = \int_K V_\alpha(x, y)\, f(y)\, d\mu(y)$$

is:

1. Hilbert-Schmidt (since $V_\alpha \in L^2(K \times K)$);

2. Self-adjoint (since V_α is real-valued and symmetric);

3. Compact (Hilbert-Schmidt implies compact);

4. Discrete-spectrum (compact self-adjoint, spectral theorem).

Hence H_α admits a complete orthonormal eigenbasis with discrete real eigenvalue spectrum $\{\lambda_k(\alpha)\}_{k=0}^{\infty}$, indexed in decreasing absolute value.

This is machine-checked in Lean 4 (file `PF/Analytic/HPGeneralOperator.lean`, theorem `H_P_at_isSelfAdjoint`).

2.2 The universal coupling

The ground state $\lambda_0(\alpha)$ of H_α is conjectured to satisfy the closed form

$$\lambda_0(\alpha) = \frac{\pi}{10\,\alpha}.$$

This conjecture is stated explicitly in Lean as the proposition `PolylogEigenvalueConjecture` (Section 5). Conditional on it, the framework yields the *universal coupling identity*:

Theorem 2.3 (Universal coupling, machine-checked)**.** *For every α_c in $\{1,\ 3/2,\ \sqrt{2},\ \varphi{+}1/4,\ 3\pi/2,\ 2,\ 3\pi/4,\ \varphi,\ \sqrt{}$*

$$\lambda_0(\alpha_c) \cdot \alpha_c = \frac{\pi}{10}$$

as an algebraic identity of the predicted closed form, irrespective of operator-theoretic considerations. Verified axiom-free in Lean (`PF/MillenniumSixReductions.lean::lambda_0_canonical_times_alpha_eq_pi_10`) and in Coq (`PF_Coq_Code/PF/QuantumGravity.v::TOE_univers`

3 Proof of Theorem 1.1 (Four-Basis Decomposition)

3.1 Numerical verification via PSLQ

The PSLQ algorithm [9] detects integer relations among a set of real numbers to a given precision. Applied to the nine α-values plus the candidate basis elements $\{1, \pi, \varphi, \sqrt{2}, \sqrt{2\pi}\}$ at 80-digit precision, PSLQ confirms the relations listed in Theorem 1.1 with residual zero (within the precision tolerance).

3.2 Algebraic verification (mechanically checked in Lean 4)

Each decomposition is also verified symbolically in Lean. The file `PF/AlphaBasisGenerators.lean` contains the following theorems, all axiom-free:

- `alpha_Poincare_from_basis`: $\alpha_{\text{Poincaré}} = 1$
- `alpha_RH_from_basis`: $\alpha_{\text{RH}} = (3/2) \cdot 1$
- `alpha_YM_from_basis`: $\alpha_{\text{YM}} = 2 \cdot 1$
- `alpha_P_from_basis`: $\alpha_P = \sqrt{2}$
- `alpha_Hodge_from_basis`: $\alpha_{\text{Hodge}} = \varphi$
- `alpha_NP_from_basis`: $\alpha_{NP} = \varphi + 1/4$
- `alpha_NS_from_basis`: $\alpha_{\text{NS}} = (3/2) \cdot \pi$

- `alpha_BSD_from_basis`: $\alpha_{\text{BSD}} = (3/4) \cdot \pi$
- `alpha_QG_from_basis`: $\alpha_{\text{QG}} = \sqrt{2} \cdot \sqrt{\pi}$

The composed statement is

Theorem 3.1 (Framework has four degrees of freedom, machine-checked)**.** *There exist real coefficients $q_1, \ldots, q_9 \in \{1, 3/2, 2, 3/4\}$ such that each α_c is the explicit algebraic combination given in Theorem 1.1. Verified in Lean as* `framework_has_four_dof` *at commit* `d8515cf`*.*

3.3 Geometric interpretation

The four-basis structure reflects three distinct geometric origins for the framework's α-values:

1. **Rational ladder** $\{\alpha_{\text{Poincaré}}, \alpha_{\text{RH}}, \alpha_{\text{YM}}\} = \{1, 3/2, 2\}$: arithmetic progression with common difference $1/2$. The midpoint identity $\alpha_{\text{RH}} = (\alpha_{\text{Poincaré}} + \alpha_{\text{YM}})/2$ holds as a machine-checked theorem.
2. **Algebraic group** $\{\sqrt{2}, \varphi, \varphi + 1/4\}$: two degree-2 algebraics over $\mathbb{Q}$. The Hodge value satisfies the golden recurrence $\alpha_{\text{Hodge}}^2 = \alpha_{\text{Hodge}} + 1$; the P-class value satisfies $\alpha_P^2 = 2$; the NP-class value shifts α_{Hodge} by $1/4$ (an integer-quadratic relation $16\alpha_{NP}^2 - 24\alpha_{NP} - 11 = 0$).
3. **π-doubling pair** $\{3\pi/4, 3\pi/2\}$: locked to the rational ladder via $\alpha_{\text{BSD}} = (\pi/2) \cdot \alpha_{\text{RH}}$ and $\alpha_{\text{NS}} = \pi \cdot \alpha_{\text{RH}}$. The doubling ratio $\alpha_{\text{NS}}/\alpha_{\text{BSD}} = 2$ is a machine-checked theorem.

The single exceptional case is $\alpha_{\text{QG}} = \sqrt{2\pi}$, the only instance coupling π into the radical group.

4 The Six Conditional Reductions

4.1 P versus NP

The Turing-machine encoding bridge

A central concern for Clay-acceptable proofs of $P \neq NP$ is the *bridge* from the standard Turing-machine definitions of P and NP (Cook 1971 [10], Karp 1972 [11]) to the framework's operator-spectral characterization. The Principia Fractalis formalization provides this bridge explicitly via a prime-power Gödel numbering of Turing-machine configurations.

In `PF/TuringEncoding/Basic.lean`, configurations of a Turing machine are encoded as natural numbers via the prime-power scheme: powers of 2 encode the machine's state, powers of 3 encode the head position, and higher primes encode the tape symbols. The encoding is the canonical Gödel numbering adapted to interface with Mathlib's `Turing.TuringMachine` library (Mathlib's `Mathlib.Computability.TuringMachine`).

In `PF/TuringEncoding/Complexity.lean`, the standard complexity classes are formalized via the standard definitions:

- `BinSymbol`: the binary alphabet $\{0, 1\}$;
- `BinString` := `List BinSymbol`: binary strings;
- `Language` := `Set BinString`: sets of binary strings, i.e., decision problems;
- Polynomial-time decidability via Mathlib's `Turing.TuringMachine` stepping;
- Classes P and NP defined as canonical sets of polynomially decidable / verifiable languages.

The encoding ι: `Language` $\to L^2(K, \mu)$ from discrete computation to the framework's continuous Hilbert space is the topic of `PF/TuringToOperator_PROOFS.lean`. The connection between standard Cook–Karp P vs NP and the framework's spectral α-classes runs through this encoding; closing the bridge formally is one component of the `MillenniumReductionSoundness` proposition (Section 5).

The conditional $P \neq NP$ theorem

The Principia Fractalis P-vs-NP reduction (Ch. 21 of [12]) introduces operators H_P and H_{NP} with canonical parameters $\alpha_P = \sqrt{2}$ and $\alpha_{NP} = \varphi + 1/4$. The conditional theorem is:

Theorem 4.1 (Conditional P $\neq$ NP, machine-checked)**.** *If* `PolylogEigenvalueConjecture` *holds, then $P \neq NP$ in the sense of [12] Definition 21.1.*

The Lean theorem is

$$\texttt{P\_NEQ\_NP : PolylogEigenvalueConjecture} \to \texttt{ClassP} \neq \texttt{ClassNP}.$$

The proof composes the algebraic theorems $\alpha_{P,NP}$-derivation, $\alpha_P \neq \alpha_{NP}$ (from $\sqrt{2} < \varphi + 1/4$), and the operator collapse hypothesis. Modulo `MillenniumReductionSoundness`*, this gives the standard Clay statement.*

4.2 Riemann Hypothesis

The framework's RH attack via the symmetrized transfer operator T_3^{sym} at $\alpha_{\text{RH}} = 3/2$ is documented in [12] Ch. 20.

Theorem 4.2 (Conditional RH, machine-checked)**.** *If* `RHSpectralSurjectivityConjecture` *holds (the surjectivity of the spectral bijection of T_3^{sym} onto the nontrivial ζ-zeros), then every nontrivial zero of $\zeta(s)$ in the critical strip satisfies* $\text{Re}(s) = 1/2$*.*

Lean theorem: `riemann_hypothesis_via_named_surjectivity`*. The Phase A inner-product hypotheses (*`hsmul_left`*,* `hsmul_right`*,* `hpos_def`*) and the Mayer 1991 §2 contractivity bound $\|T_3\| \leq 1$ are independently discharged axiom-free.*

4.3 Yang–Mills mass gap, Navier–Stokes, BSD, Hodge

Analogous conditional reductions for the remaining four problems are catalogued in `PF/MillenniumSixReducti` each depends on one or two named hypotheses from the list in Theorem 1.3. See [12] Chs. 22–25 for the manuscript content; the Lean formalization preserves the hypothesis structure while making each load-bearing claim explicit.

A particularly informative finding emerges from the Yang–Mills–Riemann-zeta connection: at $\alpha_{\text{YM}} = 2$, the fractal resonance function $R_f(\alpha, s) = \sum e^{i\pi\alpha D_3(n)}/n^s$ collapses to the Riemann zeta function:

$$R_f(2, s) = \zeta(s) \qquad \text{for } \text{Re}(s) > 1.$$

This is theorem `fractalResonance_alpha_two` in `PF/Consciousness/RfAtAlphaTwoIsZeta.lean`. The Yang–Mills mass gap and the Riemann zeta thus share the same generating structure at the algebraic level.

5 The Twelve Open Propositions

For each of the twelve named `Props` in Theorem 1.3, we document its statement and current status.

5.1 `PolylogEigenvalueConjecture`

The conjecture that the canonical α-values for the P and NP classes satisfy the algebraic equations derived from the self-adjointness of H_P and H_{NP}:

$$\alpha_P^2 = 2 \quad \wedge \quad 16\,\alpha_{NP}^2 - 24\,\alpha_{NP} - 11 = 0.$$

Status: open. The polylog spectral conjecture ([12] Ch. 21 `conj:polylog-spectrum`) is explicitly labeled as conjecture in the manuscript. Discharging this collapses the P-vs-NP reduction to unconditional.

5.2 `RHSpectralSurjectivityConjecture`

The surjectivity of the spectral bijection of T_3^{sym} onto the nontrivial ζ-zeros in the critical strip. **Status:** open. The load-bearing center of the RH attack; comparable in depth to RH itself.

5.3 The ten remaining propositions

Each of the remaining ten propositions (`Ch3LeadingOrderResonance`, `SpectralResonanceBridge`, `fractalYMMassGap`, `fractalYMRealizesContinuum`, `fractalEmergenceNoBlowup`, `BSD_equality_holds`, `fractalBSDRankEquality`, `HodgeConjecture`, `fractalHodgeConcentration`, `MillenniumReductionSoundnes` is documented at its file location in the Lean codebase.

6 Empirical Confirmation

6.1 IBM Quantum Verification dataset

The framework's α-value predictions were tested against the IBM Quantum Verification dataset of 143 mathematical problems run on quantum hardware in May 2025 [13]. The peak alpha distribution is significantly non-uniform ($\chi^2 = 75.04$ on 9 degrees of freedom, $p = 1.55 \times 10^{-12}$).

The most significant emergent cluster, identified by binomial test against the uniform null:

Theorem 6.1 (P-class empirical cluster)**.** *Of* 143 *problems tested,* 22 *exhibit peak alpha within* ± 0.05 *of* $\alpha_P = \sqrt{2} \approx 1.414$*. Under the null hypothesis of uniform distribution over the observed range, the expected count is* ≈ 7.28*. The binomial one-sided p-value is* $p \approx 3.7 \times 10^{-6}$*.*

Theorem 6.2 (P-vs-NP problem exact-prediction match)**.** *The peak alpha measured for the* P*-vs-*NP *problem itself is* 1.868*, matching the predicted* $\alpha_{NP} = \varphi + 1/4 = 1.86803398\ldots$ *to four decimal places (*$\Delta = 3.4 \times 10^{-5}$*).*

Theorem 6.3 (RH-class exact cluster)**.** *Six problems (including the Riemann Hypothesis itself, the Poincaré Conjecture, and the Closing Lemma) record peak alpha exactly* 1.500*, matching the predicted* $\alpha_{\mathrm{RH}} = 3/2$ *to the dataset's quantization precision* 0.01*.*

6.2 Independent spectral eigenvalues

Independent spectral computations on the framework's transfer operators [14] produced the following emergent eigenvalues, not hardcoded as inputs:

Computed value	Predicted λ_0	$\lvert\Delta\rvert$
0.22374	$\pi/(10\sqrt{2}) = 0.22214$	0.0016
0.22410	$\pi/(10\sqrt{2}) = 0.22214$	0.0020
0.21035	$\pi/15 = 0.20944$	0.0009

All three agree with the predicted closed forms to three decimal places, providing emergent (non-input) numerical confirmation.

7 Formal Verification Architecture

The framework is formalized in two independent proof assistants:

- **Lean 4** (with Mathlib v4.24.0-rc1): 179 source files, ≈ 1950 theorems, ≈ 150 proposition definitions, zero project axioms. Build status: 6354 jobs clean, 0 sorries.
- **Coq 8.18.0**: 31 modules covering the framework's structural core (`QuantumGravity.v`, `GeneralRelativity.v`, `FractalKernelSelfSimilarity.v`, `SpectralResonanceBridge.v`, plus the existing 24-module port).

Each non-trivial theorem in the paper is verified mechanically; the audit trail is available at [15], commit `d8515cf`.

8 Discussion

8.1 Significance of the four-basis structure

The framework's nine α-values were independently postulated, one per chapter, in [12]. Theorem 1.1 demonstrates that the nine postulates collapse to four free generators. This is the kind of structural overconstraint characteristic of correct unifying theories in physics: e.g., the Standard Model's nineteen free parameters reduce to a much smaller set in proposed grand unifications.

The four-basis decomposition is testable: any future addition to the framework (a tenth class, an extension to quantum gravity, etc.) must either reduce to the existing four generators or introduce a new generator. The current QG extension at $\alpha_{\mathrm{QG}} = \sqrt{2\pi}$ satisfies this constraint (it is the algebraic combination $\sqrt{2} \cdot \sqrt{\pi}$).

8.2 What it would take to discharge the twelve propositions

The twelve `Props` are not all of equal weight. The two load-bearing ones (`PolylogEigenvalueConjecture` and `RHSpectralSurjectivityConjecture`) are each comparable in mathematical depth to the corresponding Clay problem. The other ten are either (a) structural placeholders that should reduce to one of the load-bearing two, or (b) bridge propositions (`MillenniumReductionSoundness`, `fractalYMRealizesContinuum`) that translate the framework's internal language to standard mathematical objects.

We conjecture that a single operator-theoretic theorem, proving the universal coupling $\lambda_0(\alpha) = \pi/(10\alpha)$ as a spectral identity (not a definition) of H_α, would discharge most of the placeholder propositions simultaneously. We propose this as the highest-priority target for future work.

8.3 Honest limitations

The framework's main external citations `cohen2025*` in the bibliography point to 20 documents, of which only 2 exist as standalone artifacts in the repository (the IBM hardware dataset and the Hodge JSON). The remaining 18 are "promissory" citations to manuscripts in preparation. Resolving these into either in-paper proofs or explicit references is necessary before external review.

The IBM hardware empirical signal, while statistically significant at the $\alpha_P = \sqrt{2}$ cluster, depends on the quantum-circuit encoding choices used by the original verification team. Independent replication on a different hardware platform, or by a different team, would substantially strengthen the empirical case.

8.4 Strategic outlook

The Principia Fractalis framework, as machine-checked at commit `d8515cf`, constitutes a structural reduction of the six remaining Clay Millennium Problems to twelve named, refactorable hypotheses, organized around a four-element basis structure. We do not claim to have proved any Clay Millennium Problem unconditionally. We do claim:

1. A genuine algebraic unification: nine independently postulated α-values reduce to four generators (Theorem 1.1).

2. Empirical confirmation: statistically significant clusters in independent quantum-hardware data (Theorems 6.1, 6.2, 6.3).

3. A formally verified conditional reduction in Lean and Coq, with zero project axioms.

4. A clear strategic target: the universal-coupling spectral identity, the single theorem whose proof would substantially collapse the remaining open hypotheses.

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We thank the open mathematics community for the foundational results on which this work depends, in particular the Mayer transfer-operator program [5], the Lewis–Zagier period-function correspondence [6], the Connes noncommutative geometry approach to RH [7], and the Berry–Keating Hilbert–Pólya programme [8].

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